

Nicolas Gonzalez

SKILLS

- Programming: Python, C++/C#, SQL, Matlab, Java, OpenGL, TypeScript.
- Numerical Analysis, Scientific Computing, Machine Learning, Computer Graphics, Finite Element Analysis.
- Libraries: NumPy, Scipy, Scikit, Pandas, FFTW.
- Languages: English, Spanish, French, Portuguese.

EXPERIENCE

UCSB, Santa Barbara – Professor

June 2024 – August 2024

- Developing undergrad courses' structure and core concepts.
- Guidance, counseling and pedagogy for students.
- Developing and designing class presentation material.

UCSB, Santa Barbara – Teaching Assistant (TA)

October 2018 – September 2024

- Logistics beyond class scope (course announcements, grading, etc).
- Guidance, counseling and pedagogy for students.
- Preparing and organizing material for study/review sessions.

Uneven Labs, California – Consultant Programmer

June 2022 – September 2022

- Rule-based NFT trading (TypeScript).
- ML prediction models for NFT trading hosted on AWS (scikit).
- Extensive NFT transactions database queries for ML model training.
- Web application to query ML predictions.

QuadGraphics, Mexico City – Front-End programmer

February 2018 – July 2018

- UI debugging, maintenance and improvement (C#/JavaScript) for shipping web application.

UNAM, Mexico City – Professor

January 2017 – December 2017

- Developing undergrad courses' structure and core concepts.
- Guidance, counseling and pedagogy for students.
- Developing and designing class presentation material.

Ubisoft, Montreal – *General Programmer*

August 2013 – April 2014

- Unit testing for geometry algorithms.
- Physics dynamics integration for playable characters.
- Designing customizable interface for physics-engine parameters.

EDUCATION

UCSB, Santa Barbara, USA – *Doctor of Philosophy (PhD)*

October 2018 – June 2024

Dissertation: [A Study on Stabilized USEM for Elliptic Interface Problems and Prospects for Moving Interface](#).

Publication: “Unfitted spectral element method for interfacial models”, (with Hailong Guo and Xu Yang), Communications in Computational Physics; [arXiv:2309.17027](#) (accepted)

UNAM, Mexico City, Mexico – *Masters of Science (M.Sc.)*

January 2015 – December 2016

Thesis: [Numerics on the dissipative spin-orbit model in celestial mechanics](#).

McGill, Montreal, Canada – *Bachelor of Science (B.Sc.)*

September 2009 – April 2013

Double major in Mathematics and Computer Science.

Research Project: [Numerical study of a class of differential equations with two linearly dependent state-dependent delays](#).

AWARDS

- 2018 UCSB Graduate Fellowship (USA).
- 2015 CONACYT Graduate Scholarship (Mexico).
- 2012 ISM Undergraduate Summer Scholarship (Canada).
- 2009 Canadian National University Soccer Championship Finalist.
- 2008 Canadian National Club U-18 Soccer Championship Finalist.
- 2007 Canadian National Club U-18 Soccer Championship Finalist.